

work. Special routers called LER (Label Edge Routers) mark each packet entering the network. The network handles the packet based on the label. For example, packets marked 'urgent' will go into a high bandwidth line, while routine traffic might go to other lines. This translates into a network that is more scalable, and less complex. Thanks to the connection-less model, the service provider will be able to offer different classes of service to different users. For customers, MPLS VPN offers predictable performance without compromising on any other aspect of the network.

Who can use it?

VPN can be built over any network, even within a LAN. First, a VPN server has to be installed on a server operating system such as Windows 2000 server. Once this is done, VPN clients can be installed on any networked system.

Companies based at multiple sites can use VPN to connect all their offices. Leased lines are extremely expensive to lay and maintain, as a dedicated line has to be set up between the end points and has to be renewed annually. VPNs give a consistent throughput, as the line is used exclusively by the company, without any sharing of bandwidth.

A VPN also helps you access your company's Intranet from your home, and executives on the move can synchronise their

laptops with their office desktops. The next step in this direction is the creation of a large scale virtual organisation.

Indian VPN

All this looks good on paper, but has anyone attempted to set up a VPN?

Bharat Petroleum Corporation Limited (BPCL) wanted a low cost solution to extend ERP (SAP R/3) access to its remote locations and partners—about 130 users spread over 110 locations. VPN was chosen over VSATs and leased lines, for an obvious reason—cost! These remote locations needed connections to carry out SAP transactions, which don't require high bandwidth or an 'always on' connection. The VPN implementation started in November 2001 and was completed by March 2002. BPCL opted for the CISCO model that supports up to 1,500 simultaneous tunnels. The central gateway is located in Mumbai, and is connected to the Internet via a 2 Mbps link. Scalability is important in any retail business—new outlets are added every month and then have to be connected. With VPN, all you need to do is install a VPN client.

Popularly referred to as the Bible of cricket, Wisden offers an online version of its services as Wisden Online (www.cricinfo.com/wisendotcom/). Apart from statistics, the site offers multimedia content, competitions, etc. As both the Bangalore and London Wisden offices maintain the site, they had to be connected. Although there are

How BSNL connects India?

BSNL and MTNL offer a different breed of VPN services. They support only voice—one can use a modem to send and receive unencrypted data. In simple terms it acts as an EPBAX that's spread over large distances. It's the perfect solution for small businesses, such as travel agencies, who often need to talk to their peers in other cities. You can also use it to have cost effective chats with your girlfriends all over the country.

only two offices, the distance between them makes a leased line impossible. Wisden opted for a VPN network. The network runs on the Windows platform, using a Checkpoint firewall. The benefit of this was seen when Wisden launched a live pay-per-view video stream of England's match against India, early in 2002. The Web site got a huge response for this service, and provided a stable broadcast throughout the match, despite fluctuating demand. Part of the credit goes to their VPN network.

The Indian VPN services market is filled with many players, mostly telecom companies that set up the network and maintain it. Companies such as Cisco, Checkpoint, Microsoft and others, provide the software and hardware components used in the network. Some of the better known service providing companies are VSNL, Touchtel and Tata Teleservice. MTNL and BSNL offer only voice service.

What it all leads to

As with any technology, VPN also has its share of woes. The irony is that its strength is also its drawback—the Internet, with its unpredictable delays and the susceptibility to attack. MPLS helps one to overcome this problem, and is poised for rapid deployment by the industry—simply because, though it isn't perfect, it is still the best solution available. Due to increasing competition, companies are forced to adopt cost cutting measures. In the e-commerce era that we live in, good connectivity can make the difference between the success or failure of a business. Without a doubt, VPN is the most cost effective solution for companies to keep their employees connected.

In a secured environment, VPN is bound to become the connecting line for dispersed corporate networks. The increase in the number of vendors offering VPN is indication enough of its rising popularity. ■

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Jargon Buster

IPSec: This covers authentication and encryption of data traffic over the Internet. For IPSec to work, the sending and receiving devices must share a key. IKE protocol is a key management protocol standard which is commonly used in conjunction with the IPSec standard.

Leased Line: A communication line that is exclusively laid for the private use of a company, or an individual.

Layer 2 Forwarding (L2F): L2F was developed by Cisco, and is a layer 2 tunnelling protocol similar to PPTP.

Layer 2 Tunnelling Protocol (L2TP): L2TP can send encapsulated PPP packets over IP, x.25, frame relay or ATM networks.

Point-to-Point Tunnelling Protocol (PPTP): This is an extension of Point-to-Point protocol (PPP) built to provide remote access that can be tunnelled through the Internet. PPTP encapsulates PPP packets

using generic routing encapsulation (GRE) protocol, which gives PPTP the flexibility of handling protocols other than IP.

Tunnelling: The technique of encapsulating one protocol within another, such as IPX within IP. In the context of security, it refers to encrypting IP within IP so that the traffic may be routed securely.

VPN Concatenator: Also called VPN gateway. Data from the Internet has to enter the VPN network of a company through the VPN concatenator. It decrypts the data coming from the Internet, and encrypts the data before sending it back out over the Internet.

VPN Tunnel: This describes the connection between two or more private nodes, or LANs over a public network, typically the Internet. Encryption is used to maintain the confidentiality of the data when travelling over the Internet.